casing, the impurities in the oil from the jet get sprayed from the stationary casing due to centrifugal force. Finally the purified oil comes out through the outlet. See fig shows in 8.5.1(c) Centrifugal Type Oil Filter.

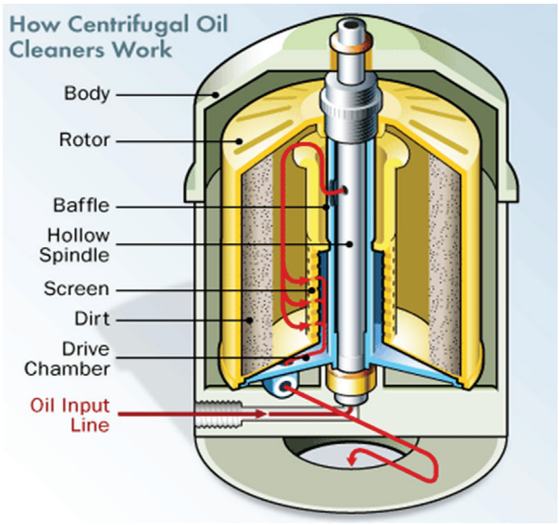


Figure 8.5.1(c) Centrifugal Type Oil Filter

8.5.2 Oil Pump

In lubrication system after the oil strainer, oil pump is placed which is the very important component of the system. Sending the oil with high pressure to the engine parts is the function of the oil pump. The oil pump is located at the crank case below the oil level. It is generally operated from the tip of distributer shaft. It gets the power from the skew gear of camshaft through distributer extension shaft. The speed of the oil pump increases with increase in the speed of the engine. This increases the pressure in the shaft. High pressure is controlled by using a pressure relief valve. The pump supplies sufficient oil to all the engine parts. Minimum pressure for the oil is maintained as 100 kPa. Normally to lubricate the engine, the lubricating oil of about 15 to 20 litres is circulated per minute. The oil is sent in large amount at high pressure through the oil pump. Hence selecting the suitable oil pump is very important. Due to bearing damage caused by friction and leakage of the oil from the engine parts there will be reduction in pressure of the oil. Hence to reduce damage of the components and to maintain sufficient pressure, suitable oil pump is needed. Figure shown in 8.5.2 Oil Pump.



Figure 8.5.2 Oil Pump

TYPES OF OIL PUMP:

The following are the types of oil pumps generally used in automobiles,

1. Gear pump.
2. Rotor pump.
3. Plunger pump.
4. Vane pump.

8.5.2.1 Gear Pump

In present automobiles, gear type oil pumps are generally used. This pump is a very simple one. It has two parts called as drive gear and driven gear. These two gears rotate together in a housing. Sufficient gap is maintained between the inner side of

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